

IMO Training – Number Theory

Valuation and Lifting the Exponent Lemma

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Introduction

Unless otherwise specified, we shall focus on integers. Furthermore, let p be an odd prime.

Definition. For $a > 0$, $\nu_p(a) = \max\{n \in \mathbb{N} : p^n \mid a\}$

Theorem. If $(x, p) = (y, p) = 1$ and $p \mid x - y$, then for all $n \in \mathbb{N}$

$$\nu_p(x^n - y^n) = \nu_p(x - y) + \nu_p(n).$$

Theorem. If $(x, p) = (y, p) = 1$ and $p \mid x + y$, then for all odd $n \in \mathbb{N}$

$$\nu_p(x^n + y^n) = \nu_p(x + y) + \nu_p(n).$$

Theorem. If $4 \mid x - y$, then for all even $n \in \mathbb{N}$

$$\nu_2(x^n - y^n) = \nu_2(x - y) + \nu_2(n).$$

Theorem. If $4 \mid x + y$, then for all even $n \in \mathbb{N}$

$$\nu_2(x^n + y^n) = \nu_2(x + y) + \nu_2(n).$$

Example. Find the order of 2 modulo 5^8 .

Example. The integer $n = 33 \cdots 3$ consists of 3^{100} copies of 3's. Find the highest power of 3 that divides n .

Example. Find the largest $n \in \mathbb{N}$ such that $5^n \mid 1001^{200!} - 101^{200!}$.

Example. Prove that there is no $a, b \in \mathbb{N}$ such that $(15a + b)(a + 15b)$ is a power of 3.

Example. Prove that $\sum_{i=1}^n \frac{1}{i}$ is not an integer for all $n \geq 2$.

Example. Prove that for primes p and $n \in \mathbb{N}$, $\nu_p(n!) = \frac{n - s_p(n)}{p - 1}$, where $s_p(n)$ is the sum of all digits of n in base p .

Example. (CGMO 2017)

- (i) Find all $n \in \mathbb{N}$ such that $4 \mid a^n - 1$ for any odd number a .
- (ii) Find all $n \in \mathbb{N}$ such that $2^{2017} \mid a^n - 1$ for any odd number a .

Example. Find all $n \in \mathbb{N}$ such that $\frac{5^n + 1}{3^n}$ is an integer.

Example. Let $a, b, c \in \mathbb{N}$ such that $c \mid a^c - b^c$. Prove that $c \mid \frac{a^c - b^c}{a - b}$.

Example. (IMO 1990) Find all $n \in \mathbb{N}$ such that $\frac{2^n + 1}{n^2}$ is an integer.

Exercises

1. Find all n such that $2^{n-1} \mid n!$.
2. Find all n such that $n \mid (n-1)!$.
3. Prove that for all $n \in \mathbb{N}$, $\frac{1}{n+1} \binom{2n}{n}$ is an integer.
4. Let $a, n \in \mathbb{N}$ such that $a^p \equiv 1 \pmod{p^n}$. Prove that $a \equiv 1 \pmod{p^{n-1}}$.
5. Find all $m, n \in \mathbb{N}$ such that $3^{m+2} \mid 4^{3^n} - 1$ and $3^{n+2} \mid 4^{3^m} - 1$.
6. (IMO 1999) Find all $x, p \in \mathbb{N}$ such that p is prime, $x \leq 2p$, and $x^{p-1} \mid (p-1)^x + 1$.

Extra Space